

EXECUTIVE INVESTMENT MEMORANDUM

PROJECT TITAN: 2-GIGAWATT AI DATA CENTER (TEXAS ERCOT REGION)

Model Architect: *Mistral Small 24B Instruct* | Benchmark Run: October 2024 | Classification: Tier-IV Mission Critical

This document provides an institutional investment analysis evaluating the capital deployment, civil engineering feasibility, power interconnection, and financial return profile for constructing a **2.0 Gigawatt AI hyperscale facility** in Texas. Based on the economic projections synthesized by **Mistral Small 24B Instruct**, total Capital Expenditure (CapEx) is modeled at **\$2,664,200,000** (Base Range: \$2,422,000,000 to \$2,664,200,000), with annualized Operational Expenditures (OpEx) of **\$100,000,000/year**. The facility is engineered to support ultra-dense AI clusters exceeding 100 kW/rack with redundant grid feeds.

Total Planned Capacity	Target Facility Area	Estimated CapEx (Mid)	Est. Annual OpEx	Target PUE
2.0 GW (2,000 MW)	1,000,000 sq ft	\$2,664,200,000	\$100,000,000	1.15 - 1.25

Capital Expenditure Allocation & Cost Centers

Item	Cost Center Description	Amount (\$USD)	Share (%)
1	Land Acquisition (50 acres)	\$10,000,000	0.4%
2	Building Shell (\$300/sq ft)	\$300,000,000	11.3%
3	Data Center Equipment (\$500/kW @ 2M kW)	\$1,000,000,000	37.5%
4	Power Infrastructure (\$300/kW @ 2M kW)	\$600,000,000	22.5%
5	Cooling Systems (\$100/kW @ 2M kW)	\$200,000,000	7.5%
6	Security & High-Speed Cabling	\$62,000,000	2.3%
7	AI Hardware & Specialized Software	\$250,000,000	9.4%
8	Contingency Reserve (10%)	\$242,200,000	9.1%
TOTAL CAPITAL INVESTMENT		\$2,664,200,000	100.0%

Strategic Architecture & Critical ERCOT Grid Risk Analysis

1. ERCOT Substation Interconnection: Interconnecting a continuous 2.0 GW load requires direct 345 kV dual-circuit transmission lines, on-site utility substations, and multi-year coordination with ERCOT grid operators. Peak summer heat and grid curtailment necessitate on-site generation and battery backup.

2. Advanced Cooling Architecture: With high-density AI accelerators exceeding 1 kW per socket, standard air cooling is insufficient. The model specifies closed-loop liquid-to-liquid heat exchangers and immersion manifolds, achieving an optimized PUE while conserving scarce regional water reserves.

3. Investment Recommendation: The economics confirm a highly defensible infrastructure asset. Initial phasing in

250 MW to 500 MW data hall increments provides de-risked capital deployment and allows forward cash flows to fund sequential phases.